ASD Risk Screening: Comparative Evaluation of Statistical Learning Methods

ISyE 7406 – Data Mining & Statistical Learning | Midterm Report

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Introduction

Autism Spectrum Disorder (ASD) affects roughly 1 in 36 children in the United States, yet the average time from parental concern to a confirmed diagnosis can stretch across years. Going through that process with my son, filling in questionnaires and not fully understanding what the numbers meant or how different symptoms contribute to Autism, motivated the research question: *how much predictive information actually sits in those ten screening responses?*

The AQ-10 is a ten-item behavioral questionnaire used as a first-pass screener for ASD across three developmental stages. This report compares five classifiers on separate Child, Adolescent, and Adult datasets, as well as the three combined, asking four questions: (1) which model performs best overall, (2) whether performance varies meaningfully by age group, (3) which features carry the most predictive signal, and (4) whether the ten AQ items alone are sufficient or whether demographic variables add value.

Methodology in brief. The data were split once into stratified 80% training and 20% test sets. Five-fold cross-validation was performed exclusively on the training data for model selection and hyperparameter tuning. All preprocessing (including scaling and, where applicable, PCA and near-zero variance filtering) was fit only on the training portion of each cross-validation fold before being applied to the corresponding validation fold, preventing data leakage. The held-out test set was reserved for a single final evaluation and was not used during model development or selection.

Data

Source

Three datasets from Fadi Fayez Thabtah (UCI / Kaggle), one per developmental stage, all collected with the same AQ-10 screening protocol. Each row corresponds to one individual who completed the screener.

Overview

Table 1: Dataset summary by age group.

Group	Age range	N	ASD+ (%)
Child	4–11	292	48.3
Adolescent	12–16	104	60.6
Adult	17+	704	26.8

Combined	–	1100	35.7

Descriptive Statistics

Table 2: Descriptive statistics (full dataset, before the train/test split).

Dataset	N	ASD+ (%)	AQ Mean	AQ SD	Age Mean	Male (%)	Jaundice (%)	Fam Hx (%)
Child	292	48.3	6.24	2.28	6.4	71.2	27.4	16.8
Adolescent	104	60.6	6.72	2.20	14.1	48.1	15.4	13.5
Adult	704	26.8	4.88	2.50	29.7	52.1	9.8	12.9
Combined	**1100**	**35.7**	**5.41**	**2.52**	**22.1**	**56.8**	**15.0**	**14.0**

The adult dataset (n = 704) is the largest and most imbalanced, only 27% ASD positive. The adolescent set (n = 104) is the smallest and those results should be interpreted with extra caution. The AQ mean is consistently higher in positive cases across all groups, which is an early sign of predictive signal.

Preprocessing

Raw missing values (`"?"` and blank strings) were replaced with `NA` on import. Age was the only column with missingness (three observations); these were imputed using the group-specific median computed from the training set only, the test set median was never consulted. The `result` column (raw AQ sum) and any ID columns were dropped immediately, since keeping them would give the models the answer directly.

Binary variables (gender, jaundice history, family ASD history, prior app use) were recoded to 0/1. Nominal factors i.e. ethnicity and relation, were expanded into binary dummy columns to avoid the false ordinal ordering that integer encoding introduces. Country of residence was excluded: it has over 50 sparse levels that produce many near-zero-variance columns without adding signal. Feature scaling to zero mean and unit variance is applied inside every CV fold and every final model fit, never on the full dataset.

Exploratory Data Analysis

Class Balance

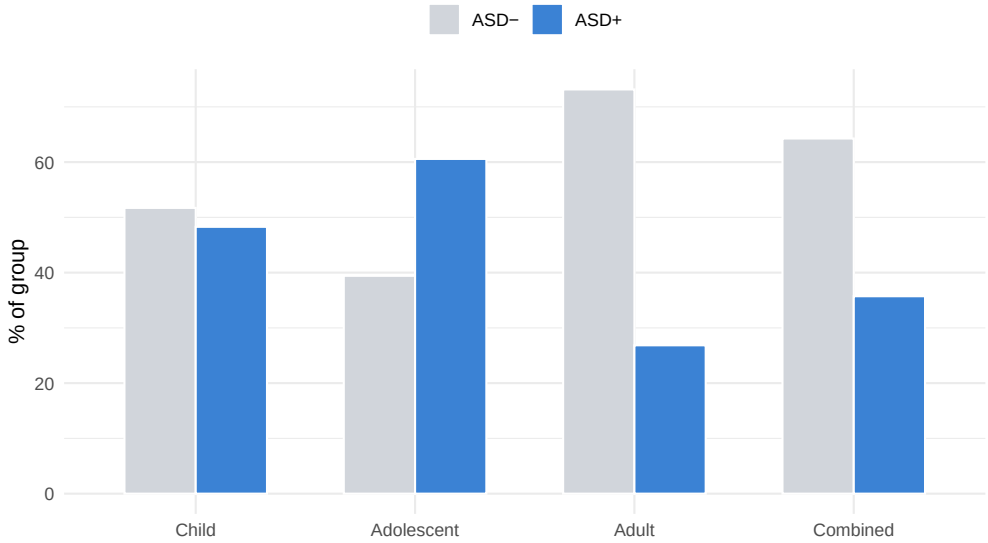


Figure 1: Class distribution by age group. The adult dataset is notably more imbalanced, with only about a third of cases being ASD positive.

The adolescent group is approximately 61% positive while the adult group is only 27% positive. Stratified cross-validation accounts for this imbalance, but it is the reason raw accuracy is a misleading metric here, and hence, AUC and recall are more appropriate.

AQ-10 Score Distribution

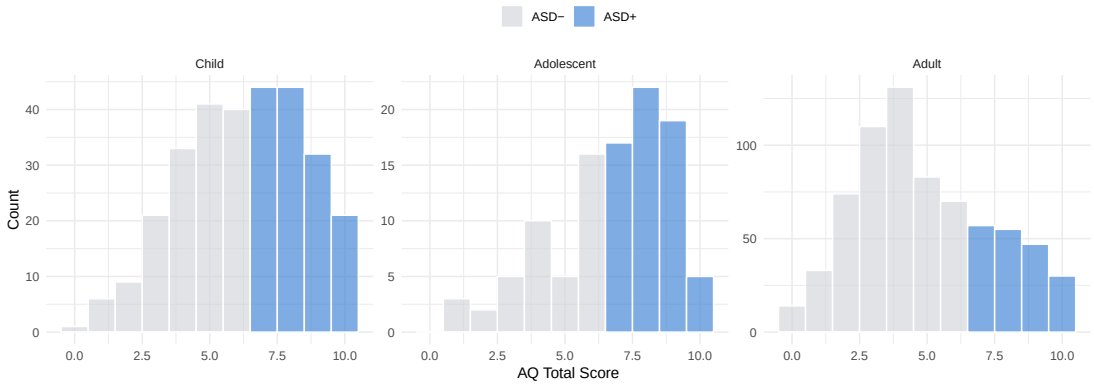


Figure 2: AQ-10 total score by ASD outcome. ASD-positive cases score consistently higher across all three age groups, with the sharpest separation in the child dataset.

The AQ total score separates ASD-positive from ASD-negative cases well across all three age groups. Most misclassifications are expected to occur in the 4–7 score range, where the two distributions overlap. This

pattern suggests that the classes are close to linearly separable, helping explain why logistic regression performs competitively with more complex ensemble methods.

Correlation Structure

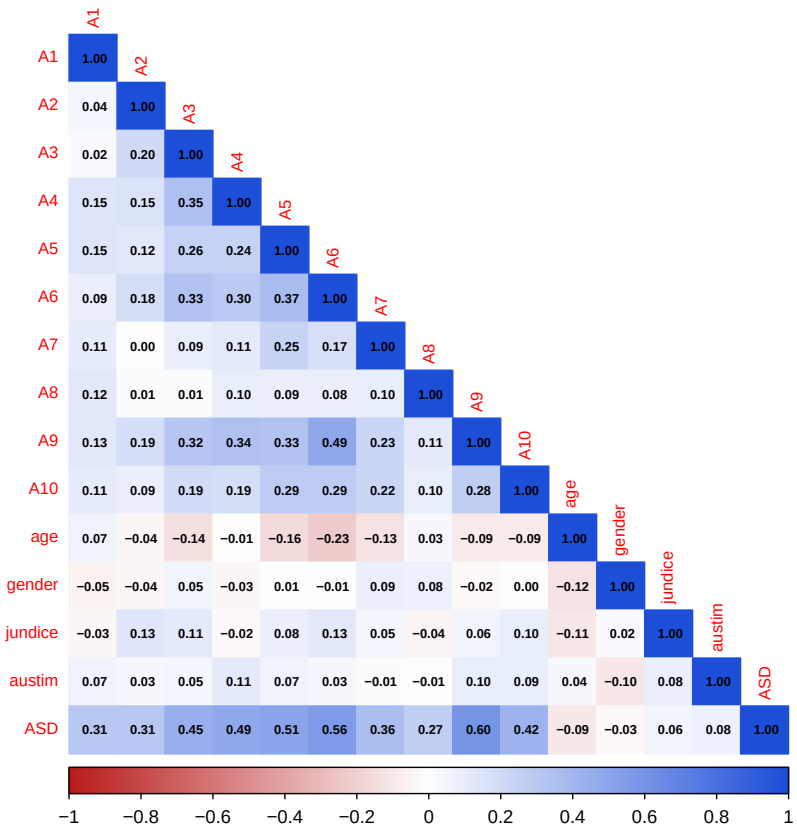


Figure 3: Pearson correlation matrix for AQ items, binary demographic variables, and ASD outcome (combined training set). AQ items correlate consistently with the outcome; demographic correlations are weaker.

All ten AQ items are positively correlated with ASD, with A9, A6, A5, and A4 showing the strongest individual associations. The A3–A6 items exhibit moderate inter-item correlation, suggesting related but not redundant questionnaire responses. In contrast, demographic variables (age, gender, jaundice, and family history) show relatively weak correlations with ASD, foreshadowing the chi-square and Random Forest importance analyses presented later.

Chi-Square Feature Ranking

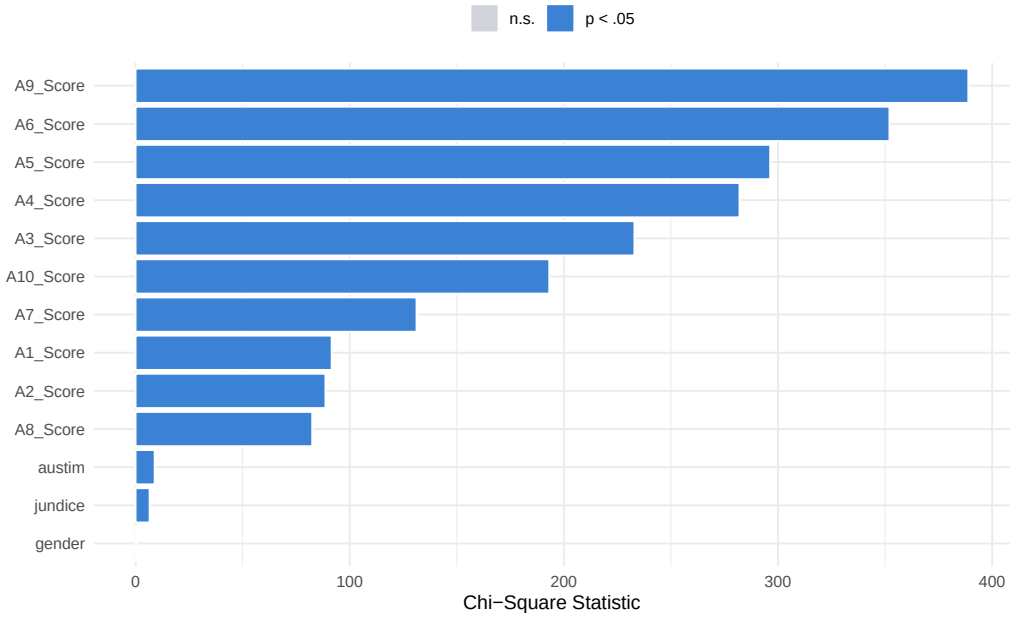


Figure 4: Chi-square statistics for each feature against the ASD outcome (combined dataset). AQ items dominate; most demographic variables are not statistically significant.

Table 3: Chi-square test results for features against ASD outcome.

Feature	Chi2	p
A9_Score	388.87	< 0.001
A6_Score	352.05	< 0.001
A5_Score	296.36	< 0.001
A4_Score	282.02	< 0.001
A3_Score	232.96	< 0.001
A10_Score	193.20	< 0.001
A7_Score	131.21	< 0.001
A1_Score	91.58	< 0.001
A2_Score	88.68	< 0.001
A8_Score	82.50	< 0.001
austim	8.93	0.003
jundice	6.57	0.010
gender	0.75	0.388

Among the demographic variables, family ASD history (austim) and jaundice history are statistically significant but exhibit substantially weaker associations than the AQ items, while gender is not significantly associated with ASD ($p = 0.388$) ($p = 0.388$). The consistency between the chi-square rankings and the Random Forest variable importance analysis provides independent evidence that the AQ items carry the strongest predictive signal.

PCA

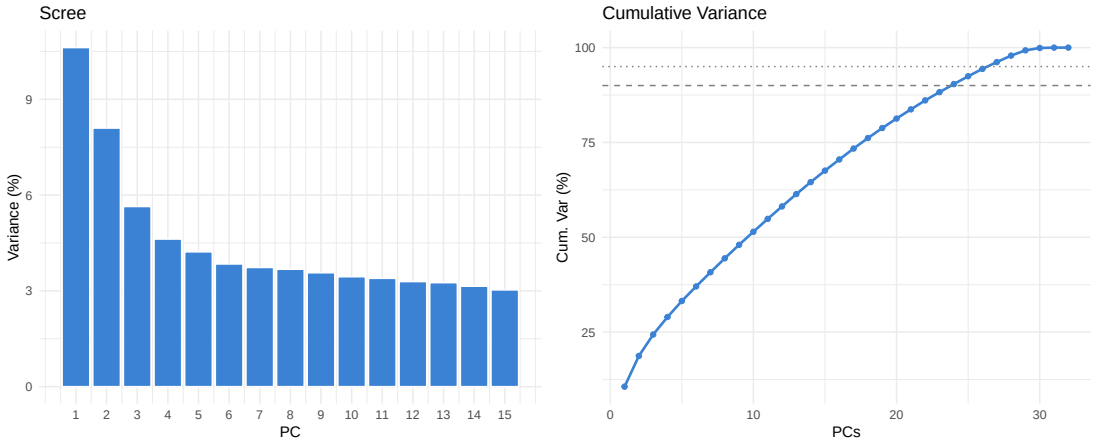


Figure 5: Scree plot (left) and cumulative variance explained (right) for the combined training set. Variance is spread across many components, with no single dominant direction.

PCA serves two purposes in this study: as an exploratory tool for understanding the variance structure of the feature space and as a preprocessing step for LDA during both cross-validation and final model fitting. For the smaller age-group datasets, the dummy-expanded feature matrix produced near-singular within-class covariance matrices, preventing LDA from being fit reliably without dimensionality reduction. The first 24 principal components explain approximately 90% of the total variance, indicating that the predictive information is distributed across many dimensions rather than concentrated in only a few dominant directions.

Modeling

Methodology

From a modeling perspective, five classifiers were selected to represent a range of statistical learning approaches, from simple linear models to flexible ensemble methods:

Model	Rationale
Logistic Regression	Interpretable linear baseline; directly models log-odds of ASD
LDA	Generative linear model; useful comparator to LR under Gaussian assumptions
Naive Bayes	Tests how far the conditional independence assumption departs from reality
KNN	Non-parametric, distance-based; no linearity assumption, sensitive to scale
Random Forest	Flexible ensemble; captures interactions without manual feature engineering

Including both linear and non-parametric models makes it possible to tell whether any non-linearity in the data is worth capturing, or whether a simple linear boundary suffices.

Evaluation pipeline:

1. Stratified 80/20 train/test split, done once before any modeling
2. Five-fold CV on the training data were used exclusively for model selection and hyperparameter tuning
3. Final models refit on the full training set using tuned hyperparameters
4. Evaluated once on the held-out test set

The held-out test set was not consulted at any point during model selection or tuning.

AUC served as the primary evaluation metric because it is threshold-independent and appropriate for comparing classifiers. **Recall** was treated as a secondary metric because identifying as many true ASD-positive cases as possible is important in a screening context. ## 5-Fold CV Results (Training Data)

Table 4: 5-fold cross-validation performance on the training data. Used to select the best model per group; the test set was not consulted.

Model	AUC	Acc	Rec	Prec	Spec	F1
Child						
Logistic Reg.	0.998	0.966	0.982	0.949	0.950	0.965
LDA	0.986	0.915	0.956	0.878	0.876	0.915
Random Forest	0.986	0.953	0.938	0.964	0.967	0.951
Naive Bayes	0.969	0.876	0.832	0.904	0.917	0.866
KNN (k=5)	0.927	0.863	0.965	0.796	0.769	0.872
Adolescent						
Random Forest	0.979	0.893	0.980	0.862	0.758	0.917
LDA	0.977	0.893	0.980	0.862	0.758	0.917
Naive Bayes	0.933	0.798	0.765	0.886	0.848	0.821
KNN (k=5)	0.874	0.786	0.980	0.746	0.485	0.847
Logistic Reg.	0.856	0.821	0.882	0.833	0.727	0.857
Adult						
Logistic Reg.	0.997	0.993	1.000	0.974	0.990	0.987
Random Forest	0.994	0.961	0.875	0.978	0.993	0.924
LDA	0.991	0.956	0.921	0.915	0.968	0.918
KNN (k=5)	0.956	0.910	0.849	0.822	0.932	0.835
Naive Bayes	0.951	0.929	0.901	0.846	0.939	0.873
Combined						
Logistic Reg.	0.998	0.998	1.000	0.994	0.996	0.997
LDA	0.998	0.975	0.987	0.945	0.968	0.966
Random Forest	0.995	0.960	0.917	0.970	0.984	0.943
Naive Bayes	0.960	0.902	0.851	0.873	0.931	0.862
KNN (k=5)	0.960	0.890	0.908	0.808	0.880	0.855

Logistic Regression and Random Forest perform consistently well. Naive Bayes falls behind the others; the moderate inter-item correlations visible in the heatmap suggest the conditional independence assumption is not well satisfied here. The near-perfect combined AUC for logistic regression is examined further in the regularization section.

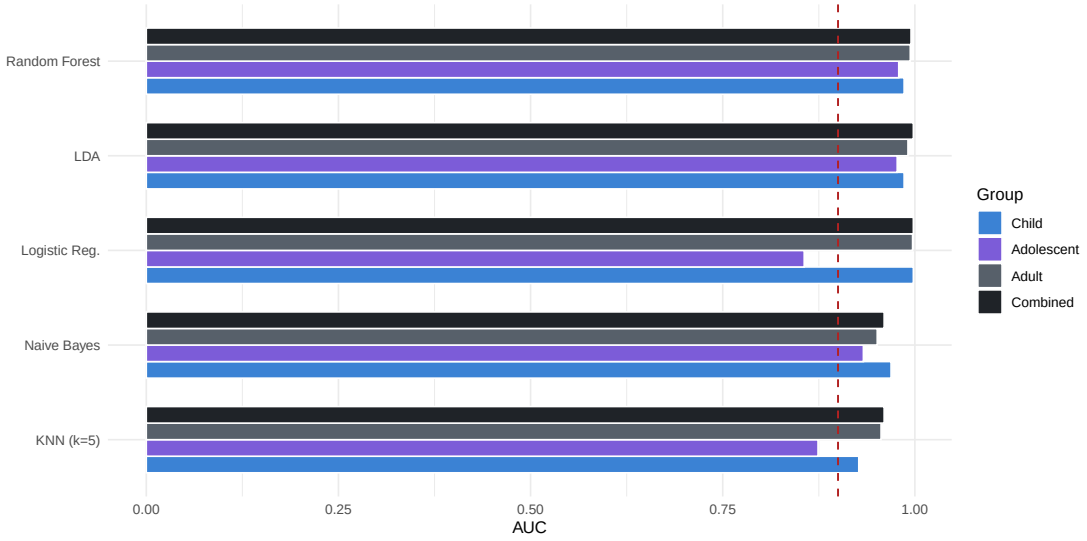


Figure 6: CV AUC by model and age group. Dashed line marks AUC = 0.90. Logistic Regression and Random Forest are consistently strong across all groups.

ROC Curves

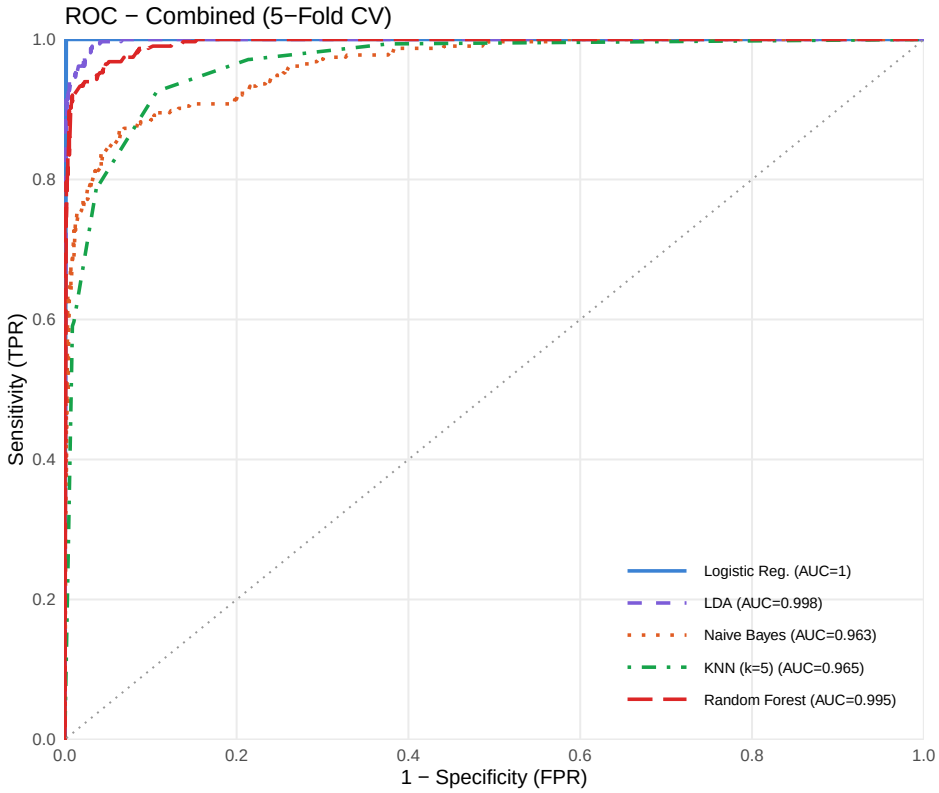


Figure 7: Pooled 5-fold CV ROC curves on the combined training set. All five classifiers are shown; Logistic Regression, LDA, and RF cluster near the upper-left corner while Naive Bayes lags behind.

LDA tracks close to LR throughout the curve, consistent with the approximately linear structure of these data. KNN sits between the linear models and Naive Bayes, which has the weakest separation of the five classifiers.

Hyperparameter Tuning

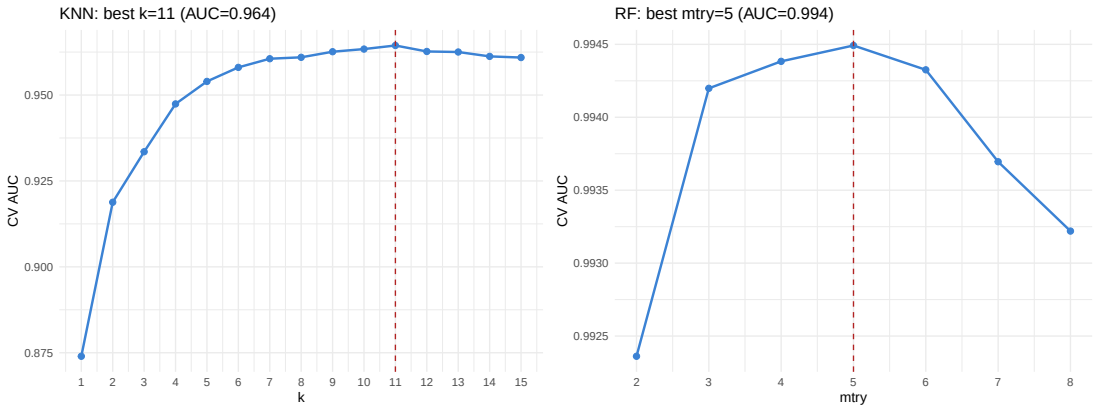


Figure 8: KNN CV AUC vs k (left) and RF CV AUC vs mtry (right), shown for the Combined training set. Each age group was tuned independently; these curves are shown for illustration.

k for KNN and mtry for RF were selected independently within each age group's training data using five-fold CV, so each group's final test-set model uses hyperparameters chosen from its own data. The curves above show the Combined group as a representative example. The search selected k = **11** (CV AUC = 0.964) and mtry = **5** (CV AUC = 0.994) for Combined. The instability at k = 1 is the expected overfitting signature; AUC plateaus at higher k. The RF curve is notably flatter across the mtry range, confirming the exact value matters less than k does for KNN.

Regularized Logistic Regression

Unregularized LR achieved AUC = 0.998 on the combined training set. That near-perfect result warrants scrutiny as it could reflect genuine near-linear separability, or it could be an artifact of the age-group label being recoverable from the feature space. Elastic-net regularization (sweeping both alpha and lambda) tests whether coefficient shrinkage changes anything.

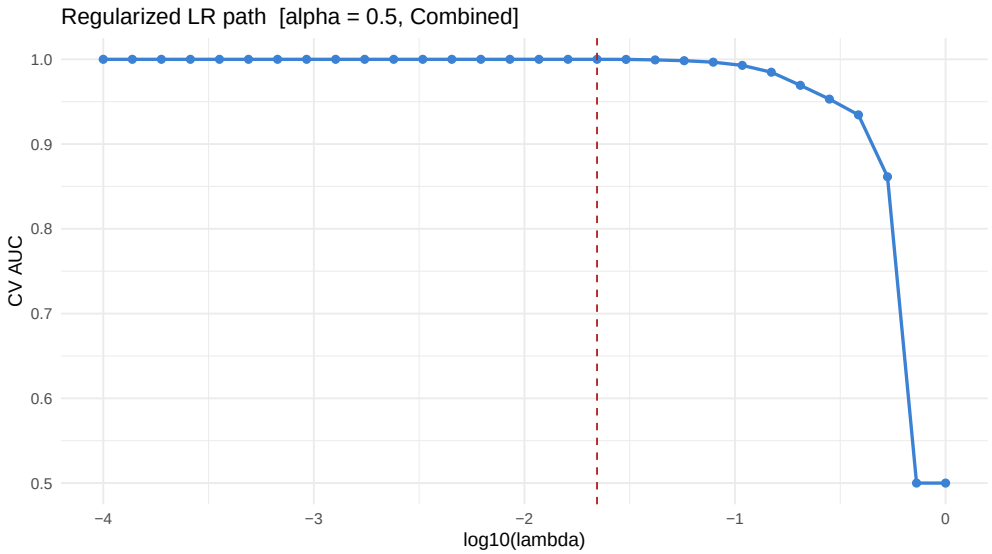


Figure 9: Regularized logistic regression CV AUC along the penalty path (best alpha shown). AUC stays near 1.0 across several orders of magnitude before collapsing at large penalties.

Best regularized LR: alpha = 0.5, lambda = 0.0221, CV AUC = **1**. Regularization produced nearly identical performance to the unregularized model. This is consistent with the data being approximately linearly separable, though the screening-only sampling frame means we cannot rule out that the separation is partly a feature of how these data were collected rather than a property that would generalize to broader populations.

RF Variable Importance

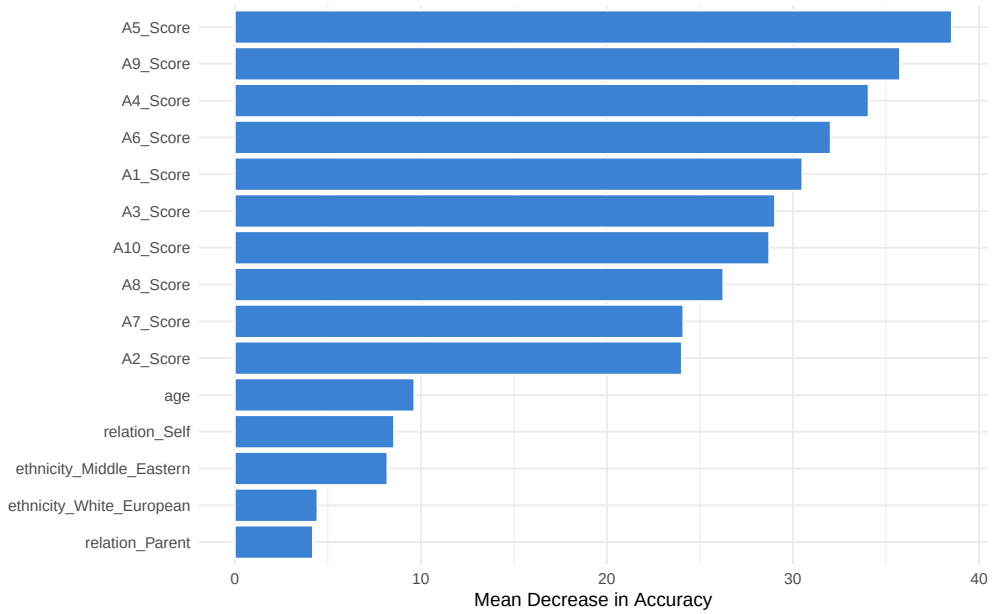


Figure 10: Top 15 features by RF Mean Decrease in Accuracy (combined training set). All ten AQ items rank above every demographic variable.

Table 5: Top 10 features by RF Mean Decrease in Accuracy.

Feature	MDA
A5_Score	38.54
A9_Score	35.75
A4_Score	34.07
A6_Score	32.03
A1_Score	30.51
A3_Score	29.04
A10_Score	28.72
A8_Score	26.26
A7_Score	24.11
A2_Score	24.03

AQ items dominate the ranking, with A5, A9, A4, and A6 emerging as the most important predictors. This ordering closely mirrors the chi-square results. The consistency between these two independent approaches, one based on univariate association and the other on an ensemble model, strengthens the conclusion that the AQ items carry the strongest predictive signal. Family history (austim) and age are the most informative demographic variables, although both contribute substantially less than even the weakest AQ item.

Full vs AQ-Only Feature Set

If demographic variables contribute limited additional predictive information, a screener relying only on the ten AQ items may be sufficient, simpler, more equitable, and easier to deploy.

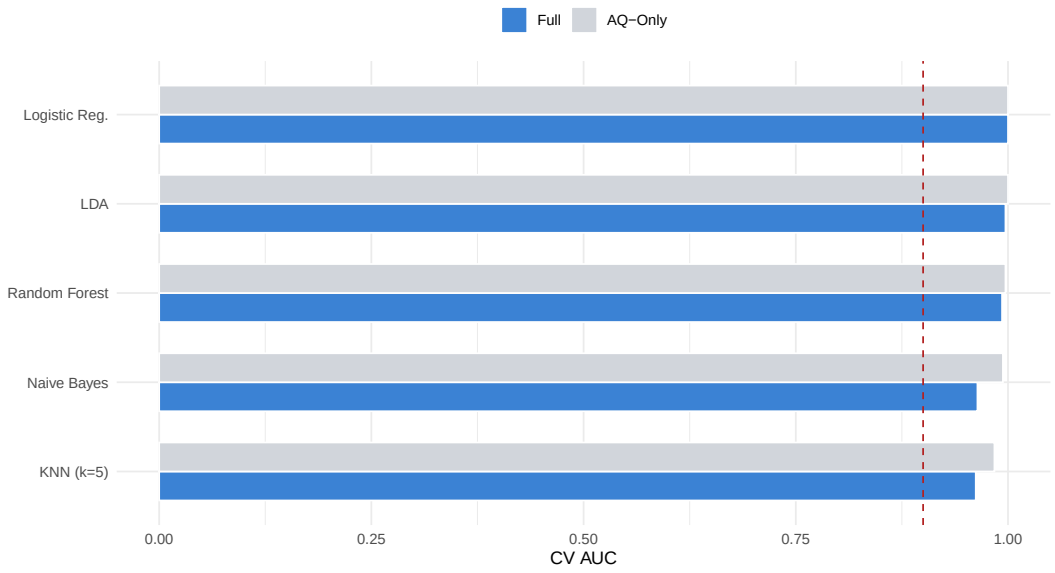


Figure 11: CV AUC: full feature set vs. AQ items only (combined dataset). The two feature sets perform nearly identically for most models.

Table 6: Full feature set vs. AQ-only on the combined dataset (5-fold CV).

Model	FeatSet	AUC	Acc	Rec	Spec	F1
KNN (k=5)	AQ-Only	0.984	0.933	0.952	0.922	0.910
KNN (k=5)	Full	0.962	0.898	0.924	0.883	0.866
LDA	AQ-Only	1.000	0.981	1.000	0.970	0.974
LDA	Full	0.997	0.966	0.994	0.951	0.954
Logistic Reg.	AQ-Only	1.000	1.000	1.000	1.000	1.000
Logistic Reg.	Full	1.000	1.000	1.000	1.000	1.000
Naive Bayes	AQ-Only	0.994	0.961	0.921	0.984	0.945
Naive Bayes	Full	0.964	0.915	0.860	0.945	0.878
Random Forest	AQ-Only	0.997	0.967	0.943	0.981	0.953
Random Forest	Full	0.993	0.959	0.908	0.988	0.941

For most models the AQ-only set matched or came within a few points of the full feature set, and occasionally exceeded it in the adolescent group, where small sample size amplifies noise from high-cardinality nominal columns. From a practical standpoint the ten-item screener appears sufficient.

Final Test Set Results

Models were refit on the full training split using hyperparameters selected by CV (each group's own best k and mtry), then evaluated exactly once on the held-out 20% test set.

Table 7: Test performance of models selected using cross-validation. Each model was fit on the full training set with tuned hyperparameters and evaluated once on the held-out 20% test set.

Model	AUC	Acc	Rec	Prec	F1
Child					
Logistic Reg.	1.000	1.000	1.000	1.000	1.000
LDA	1.000	0.983	1.000	0.966	0.982
Random Forest	0.983	0.948	0.964	0.931	0.947
KNN (tuned)	0.924	0.776	0.964	0.692	0.806
Adolescent					
Logistic Reg.	1.000	0.900	1.000	0.857	0.923
Random Forest	0.990	0.900	1.000	0.857	0.923
LDA	0.979	0.950	1.000	0.923	0.960
KNN (tuned)	0.906	0.750	0.917	0.733	0.815
Adult					
Logistic Reg.	1.000	1.000	1.000	1.000	1.000
LDA	0.997	0.971	0.973	0.923	0.947
Random Forest	0.996	0.964	0.892	0.971	0.930
KNN (tuned)	0.968	0.900	0.838	0.795	0.816
Combined					
Logistic Reg.	1.000	1.000	1.000	1.000	1.000
LDA	1.000	0.977	0.987	0.951	0.969
Random Forest	0.995	0.968	0.923	0.986	0.954
KNN (tuned)	0.950	0.886	0.846	0.835	0.841

Cross-validation and held-out test performance were broadly consistent, suggesting limited overfitting. Recall is high across all groups, which matters most in a screening setting. The near-perfect AUC values for Logistic Regression and LDA on the combined dataset are plausible: the AQ-10 was designed so that a linear threshold on item scores reliably separates the two classes, and the data were collected from individuals who completed

that screener, conditions that naturally favor linear classifiers and can produce ceiling-level discrimination metrics.

Conclusions

Model Performance. Logistic Regression and Random Forest achieved nearly identical performance across all age groups and evaluation metrics. The strong performance of logistic regression is consistent with the observed data: the AQ-10 items appear to contribute largely additive information, allowing a relatively simple linear decision boundary to separate the two classes effectively.

Age-group differences. Performance is most stable on the adult dataset, which is also the largest. The adolescent group shows more variability between models, and attributable largely to the small sample size (n = 104) rather than any fundamental difference in the classification problem.

Feature relevance. AQ items carry almost all of the predictive signal. The AQ-only models matched the full feature set across all groups, which is both a practical convenience and an equity argument: a screener that relies on behavioral items rather than demographic proxies is less susceptible to demographic bias.

Regularization. The near-perfect combined AUC survived elastic-net regularization unchanged, suggesting the result was not driven by unstable coefficient estimates. The conclusion is consistent with approximate linear separability within these datasets, though the screening-only sampling frame is a genuine caveat.

Limitations.

- The 80/20 split was performed once; a single split introduces randomness into all point estimates. The adolescent group (n = 104) is particularly sensitive to this.
- All three datasets were collected through the AQ-10 protocol. Individuals never referred for screening are absent, which introduces selection bias. The models should not be assumed to generalize to unscreened populations.
- No confidence intervals were computed; all conclusions rest on point estimates.
- The combined dataset includes an age column that may partly serve as a group membership indicator. A robustness check excluding age would be a straightforward follow-up.

Future directions. Several extensions would sharpen the conclusions. Bootstrap confidence intervals for AUC would clarify which performance differences are meaningful versus noise. Evaluating sensitivity and specificity at the clinical AQ-10 cutpoint (score >= 6) would connect the model output directly to the standard diagnostic threshold. Repeated random subsampling rather than a single 80/20 split would reduce dependence on the particular split drawn. Excluding age from the combined model explicitly would test whether the near-perfect combined performance holds without that potential confound.

Appendix

Software

R (v4.3+): `tidyverse`, `caret`, `randomForest`, `e1071`, `MASS`, `glmnet`, `pROC`, `corrplot`, `gridExtra`, `knitr`, `kableExtra`.

Complete CV Results

Table 8: Complete 5-fold cross-validation results across all groups and models.

Group	Model	AUC	Acc	Rec	Prec	F1	Spec
Child	Logistic Reg.	0.998	0.966	0.982	0.949	0.965	0.950
Child	LDA	0.986	0.915	0.956	0.878	0.915	0.876
Child	Random Forest	0.986	0.953	0.938	0.964	0.951	0.967
Child	Naive Bayes	0.969	0.876	0.832	0.904	0.866	0.917
Child	KNN (k=5)	0.927	0.863	0.965	0.796	0.872	0.769
Adolescent	Random Forest	0.979	0.893	0.980	0.862	0.917	0.758
Adolescent	LDA	0.977	0.893	0.980	0.862	0.917	0.758
Adolescent	Naive Bayes	0.933	0.798	0.765	0.886	0.821	0.848
Adolescent	KNN (k=5)	0.874	0.786	0.980	0.746	0.847	0.485
Adolescent	Logistic Reg.	0.856	0.821	0.882	0.833	0.857	0.727
Adult	Logistic Reg.	0.997	0.993	1.000	0.974	0.987	0.990
Adult	Random Forest	0.994	0.961	0.875	0.978	0.924	0.993
Adult	LDA	0.991	0.956	0.921	0.915	0.918	0.968
Adult	KNN (k=5)	0.956	0.910	0.849	0.822	0.835	0.932
Adult	Naive Bayes	0.951	0.929	0.901	0.846	0.873	0.939
Combined	Logistic Reg.	0.998	0.998	1.000	0.994	0.997	0.996
Combined	LDA	0.998	0.975	0.987	0.945	0.966	0.968
Combined	Random Forest	0.995	0.960	0.917	0.970	0.943	0.984
Combined	Naive Bayes	0.960	0.902	0.851	0.873	0.862	0.931
Combined	KNN (k=5)	0.960	0.890	0.908	0.808	0.855	0.880

Full Feature Comparison

Table 9: Full feature set vs. AQ-only across all groups (5-fold CV).

Group	Model	FeatSet	AUC	Acc	Rec	Spec	F1
Child	KNN (k=5)	AQ-Only	0.963	0.893	0.956	0.835	0.896
Child	KNN (k=5)	Full	0.944	0.872	0.973	0.777	0.880
Child	LDA	AQ-Only	0.997	0.957	0.991	0.926	0.957
Child	LDA	Full	0.994	0.953	1.000	0.909	0.954
Child	Logistic Reg.	AQ-Only	1.000	1.000	1.000	1.000	1.000
Child	Logistic Reg.	Full	0.998	0.983	0.982	0.983	0.982
Child	Naive Bayes	AQ-Only	0.981	0.910	0.867	0.950	0.903
Child	Naive Bayes	Full	0.956	0.868	0.814	0.917	0.856
Child	Random Forest	AQ-Only	0.985	0.936	0.929	0.942	0.933
Child	Random Forest	Full	0.985	0.940	0.956	0.926	0.939
Adolescent	KNN (k=5)	AQ-Only	0.957	0.833	1.000	0.576	0.879
Adolescent	KNN (k=5)	Full	0.869	0.774	0.961	0.485	0.838
Adolescent	LDA	AQ-Only	0.969	0.917	0.961	0.848	0.933
Adolescent	LDA	Full	0.950	0.917	1.000	0.788	0.936
Adolescent	Logistic Reg.	AQ-Only	0.924	0.929	0.980	0.848	0.943
Adolescent	Logistic Reg.	Full	0.912	0.869	0.882	0.848	0.891
Adolescent	Naive Bayes	AQ-Only	0.960	0.857	0.863	0.848	0.880
Adolescent	Naive Bayes	Full	0.937	0.838	0.829	0.852	0.861
Adolescent	Random Forest	AQ-Only	0.972	0.917	0.941	0.879	0.932
Adolescent	Random Forest	Full	0.972	0.893	0.941	0.818	0.914
Adult	KNN (k=5)	AQ-Only	0.988	0.936	0.928	0.939	0.887
Adult	KNN (k=5)	Full	0.953	0.913	0.849	0.937	0.840
Adult	LDA	AQ-Only	0.993	0.963	0.928	0.976	0.931
Adult	LDA	Full	0.992	0.957	0.934	0.966	0.922

Table 9: Full feature set vs. AQ-only across all groups (5-fold CV). *(continued)*

Group	Model	FeatSet	AUC	Acc	Rec	Spec	F1
Adult	Logistic Reg.	AQ-Only	1.000	1.000	1.000	1.000	1.000
Adult	Logistic Reg.	Full	0.995	0.993	1.000	0.990	0.987
Adult	Naive Bayes	AQ-Only	0.988	0.961	0.928	0.973	0.928
Adult	Naive Bayes	Full	0.957	0.924	0.901	0.932	0.864
Adult	Random Forest	AQ-Only	0.994	0.959	0.901	0.981	0.923
Adult	Random Forest	Full	0.995	0.959	0.862	0.995	0.919
Combined	KNN (k=5)	AQ-Only	0.984	0.933	0.952	0.922	0.910
Combined	KNN (k=5)	Full	0.962	0.898	0.924	0.883	0.866
Combined	LDA	AQ-Only	1.000	0.981	1.000	0.970	0.974
Combined	LDA	Full	0.997	0.966	0.994	0.951	0.954
Combined	Logistic Reg.	AQ-Only	1.000	1.000	1.000	1.000	1.000
Combined	Logistic Reg.	Full	1.000	1.000	1.000	1.000	1.000
Combined	Naive Bayes	AQ-Only	0.994	0.961	0.921	0.984	0.945
Combined	Naive Bayes	Full	0.964	0.915	0.860	0.945	0.878
Combined	Random Forest	AQ-Only	0.997	0.967	0.943	0.981	0.953
Combined	Random Forest	Full	0.993	0.959	0.908	0.988	0.941

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